Neighborhood Income and Crime Density in Toronto

AZEEZ ABIDEMI ADIGUN

HLSC 3Q20

BROCK UNIVERISTY

DR. SAMANTHA HAJNA

INTRODUCTION

Crime and violence are major public health issues because they affect how people live, feel, and connect with their communities. Many studies show that being exposed to crime or violence is strongly linked to negative health problems, including higher rates of PTSD, anxiety, depression, and other mental health challenges (Cisler et al., 2012, Fowler et al., 2009). These effects appear even when people are not directly harmed, because the stress of living in a high crime level neighbourhood still affects the body and they may go outside less, interact less with neighbors, or feel cut off from others, and these changes can lead to worse physical and mental health (Foster & Giles-Corti, 2008). The constant stress that comes with living in unsafe or violent environments can also impact the body by increasing the risk of heart disease and other chronic conditions over time (Sundquist et al., 2006). For children and teenagers, living around violence has been connected to problems with learning, lower academic performance and long-term mental health struggles (Sharkey et al., 2012).

The link between socioeconomic status and crime has been shown across many years of research. Neighborhoods with lower incomes usually experience higher levels of both violent and property crime compared to wealthier areas (Sampson et al., 1997). This pattern appears because of many factors, such as poverty, weak social connections, fewer economic opportunities, limited community resources and overall social disadvantage (Bursik & Grasmick, 1993). This unequal spread of crime in poorer neighborhoods is a form of environmental injustice because it exposes the most vulnerable people to harmful conditions that can negatively affect health (Kawachi et al., 1999).Geographic information systems, commonly known as GIS, are powerful tools for studying how crime appears across different places (Chainey & Ratcliffe, 2005). GIS makes it possible to examine high crime zones, how crime compares across neighborhoods, and where public health efforts should be directed. In a large, diverse city like Toronto, understanding how crime and income vary across neighborhoods is important for building safer communities and improving health equity.

The purpose of this review was to study how median household income and major crime density are spread across Toronto's 158 official neighborhoods using 2021 Census income data and 2023 crime records. The study focused on creating well-detailed maps that show neighborhood patterns and identify where public health attention may be needed most. This matters because spatial analysis of crime helps us guide public health planning, supports decisions about where violence prevention programs should go, helps explain how social conditions affect health inequities and guide equal distribution of the city resources among all neighborhood

METHODS

Data Sources

Three main data sources were used. The first source was the neighborhood boundary file that outlines Toronto's 158 official neighborhoods, which came from the City of Toronto Open Data portal (City of Toronto, 2021). The second source was median household income from the 2021 Census, which was accessed through the City of Toronto Open Data portal (Statistics Canada, 2022). This data shows median after-tax household income for the 2020 tax year. The third source was major crime incidents reported in 2023, downloaded from the Toronto Police Service Public Safety Data Portal (Toronto Police Service, 2023). This dataset included all reported assaults, robberies, break and enters, auto thefts, and thefts over $5,000. Each incident had information such as the date, type of crime, and geographic coordinates. Altogether, the dataset included all the crime incidents across Toronto for the year.

Data Management

The city boundary data was downloaded as a shapefile, The income data was downloaded as a CSV file and cleaned in Microsoft Excel. The original Census dataset had many variables, so only median after-tax income and neighborhood names and ID were kept for this analysis. Neighborhood names was checked and adjusted to make sure they matched the names in the neighborhood boundary file. Seven neighborhoods did not have income data, likely because of confidentiality rules for small populations (Statistics Canada, 2022). These neighborhoods were still included for crime analysis but removed from income comparisons. The crime data was downloaded as a CSV file containing crime records from 2014 to 2024 and imported into ArcGIS Pro. The dataset was filtered using the "Select by Attributes" tool to filter all crimes in 2023. All 49,432 crime incidents included coordinates within Toronto.

Geospatial Analysis

All GIS steps were completed using ArcGIS Pro. First, the cleaned income file was converted into a dbf format and joined to the boundary file using neighborhood names as the matching field. The final joined file was saved as a new feature file. Next, crime coordinates were converted into points using the Display XY Data tool. These points aligned properly with Toronto's neighborhood boundaries. A spatial join was used to count how many crime incidents occurred in each neighborhood. Neighborhood polygons were the target layer and crime points were the join layer. The "Intersect" method was used so that any crime point located inside a neighborhood would be counted. The new file "Join_Count" created from this join included a field that showed the total crime incidents

for each neighborhood. Then, neighborhood area was calculated in square kilometers using geodesic. Crime density was then calculated using the formula Crime_Density = Join_Count / Area_km^2.

Map Illustration

Two choropleth maps were created to show the main variables. Map 1 displayed median household income, and Map 2 displayed crime density. Both maps used five categories created using the Natural Breaks (Jenks) method, which created categories that highlights the differences across the data Map 1 used a yellow-to-red color scale, where lighter colors showed lower income and darker colors showed higher income. Map 2 used a light blue- to - dark blue scale, where lighter shades showed lower crime density and darker shades showed higher crime density. This was done to show visual comparison between income and crime patterns without color confusion. Both maps included required cartographic elements such as titles, legends, north arrows, scale bars, source citations, author name and date. Maps were exported as JPEG files.

RESULTS

Data Summary

Variable	N	Minimum	Maximum	Mean	Median	Standard Deviation
Median Household Income ($)	151	27,200	61,600	37,821	35,600	7,765
Total crime count	158	74	1462	309	251.5	206
Crime Density (per km^2)	158	12.02	1421.79	127.89	78.18	179.34

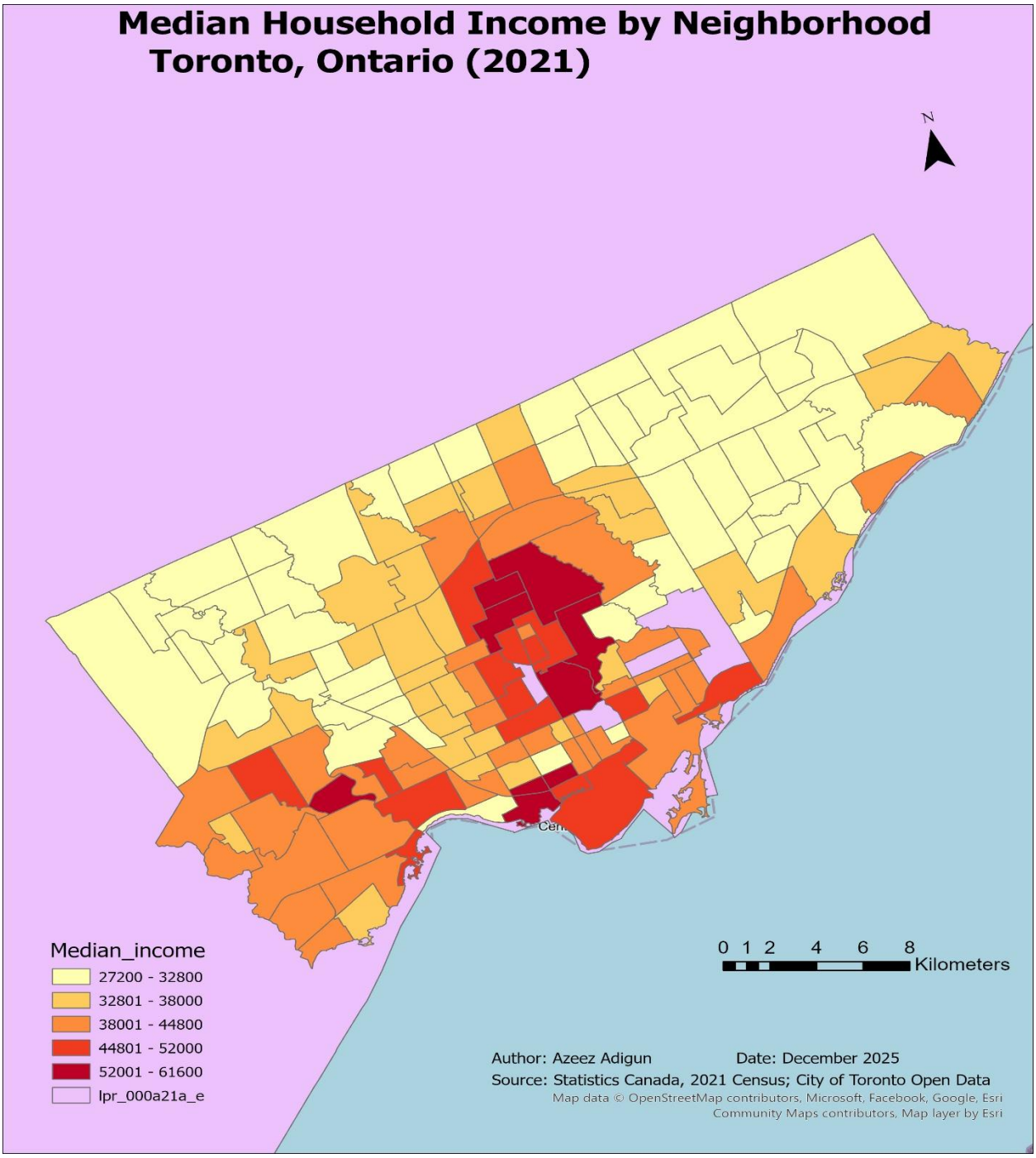
Table 1

Table 1 shows the data summary for median household income, total crime counts and crime density across all neighborhoods. Income data was available for 151 neighborhoods. Income ranged from $27,200 to $61,600, with a mean of $37,821 and a median of $35,600. This shows a wide spread in income levels.

Total crime counts per neighborhood ranged from 74 to 1,462 incidents. The mean crime count 309.1 incidents was higher than the median count 251.5 incidents, indicating a strong concentration of crime in minority of neighborhoods. Crime density ranged from

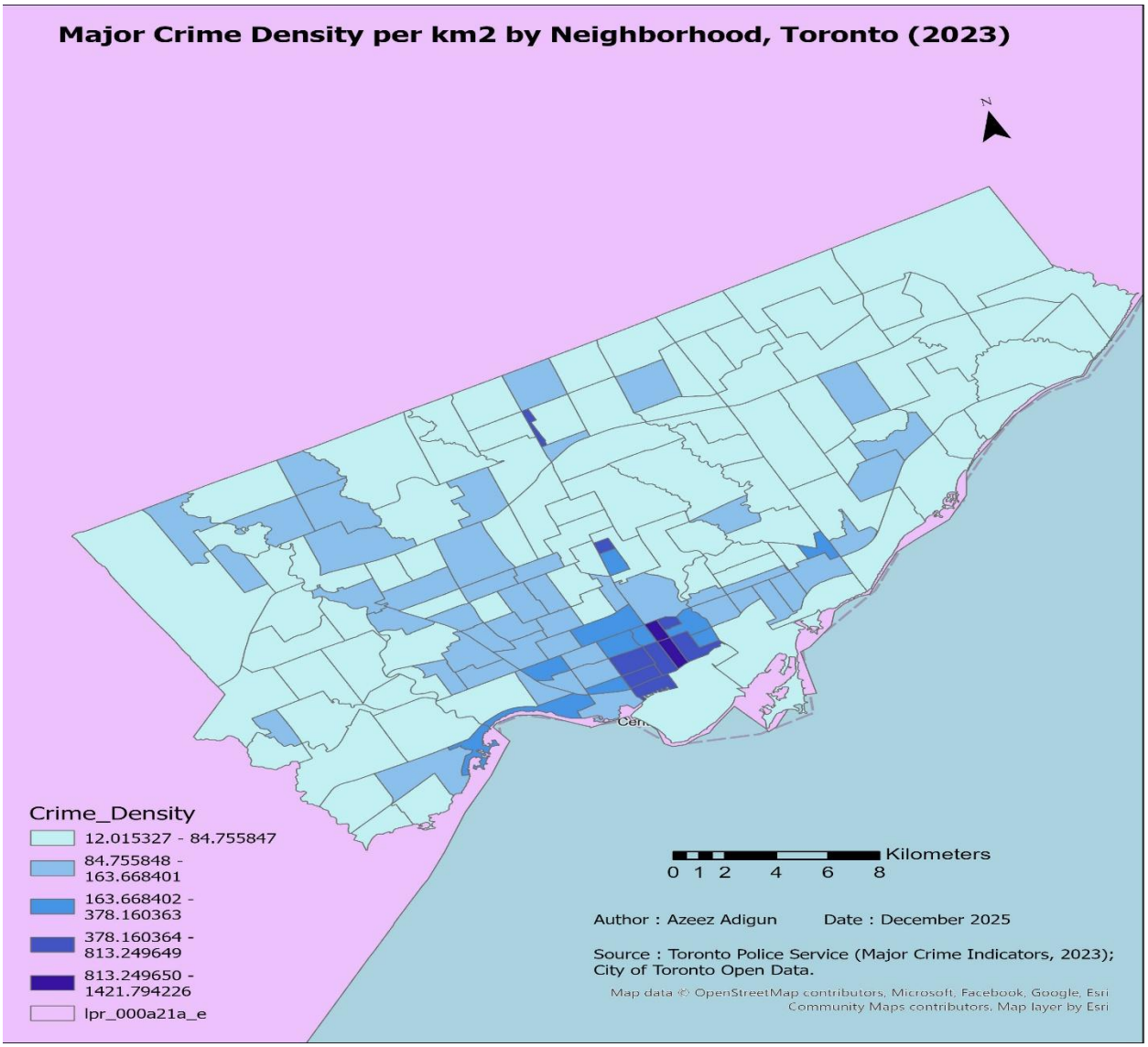
12.02 to 1,421.79 crimes per km^2, with a mean of 127.89 and a median of 78.17. This shows that crime is spread unevenly across the city.

Map of Median Household Income



Map 1 shows how income levels in households are spread across Toronto neighborhoods. There are clear patterns visible. The richest neighborhoods (over $52,000) are mostly at the top of the map, grouped together as high-income areas. The poorest neighborhoods (under $32,800) are spread across the middle of the map. Downtown, in the lower middle part, has both rich and poor areas, so it looks mixed up instead of even.

Map of Crime Density



Map 2 shows a very different pattern for crime density. The most noticeable thing is the extreme concentration of crime in downtown which is visible in the lower central portion of the map. The highest crime densities (over 378 crimes/km^2) are tightly focused in this zone, with an even smaller area exceeding 813 crimes/km^2, showing the city's most dangerous area. Neighborhoods farther from downtown, especially along the edges of the map, have

much lower crime densities (under 85 crimes/km^2) and appear as the lightest-colored zones which represents the safest areas. The middle areas between downtown and the edge show moderate crime densities (163–378 crimes/km^2).

Comparing the two maps shows that crime density does not directly correspond to income patterns. While the wealthiest areas in the upper portions generally have low crime, the lowest-income neighborhoods in the middle belt do not have the highest crime. Instead, the most crime occurs downtown, where incomes are mixed. This suggests that crime is influenced by many factors beyond income, such as commercial activity, population density, land use and access to transit.

DISCUSSION

This analysis showed clear differences in how income and crime density are spread across Toronto. Income levels in the highest neighborhoods were almost three times those in the lowest neighborhoods. Crime density varied by much more, showing huge differences across the city. The highest crime density appeared in downtown, while the lowest values were found in the outer suburbs.

These results support previous research demonstrating that crime is often higher in urban areas. (Sampson et al., 1997). Theories in criminology explain how neighborhood factors such as poverty, instability, and weak community structures relate to higher crime levels (Bursik & Grasmick, 1993). Other studies in Canada also show that crime is more prevalent in city centers and neighborhoods facing socioeconomic hardship. (Silver, 2011). High crime levels in downtown Toronto may be influenced by factors such as high population density, busy commercial areas, nightlife activity, and major transportation routes (Andresen, 2006; Weisburd, 2015). Crime is also commonly found in focused local areas resulting from highly specific environmental or social factors. (Weisburd, 2015).

From a public health view, these results point to serious health equity concerns because exposure to crime acts as a constant environmental stressor (Mair et al., 2010, Lorenc et al., 2012). Research also shows that fear of crime can reduce physical activity and worsen mental health (Lorenc et al., 2012). Studies conducted in Toronto indicate that disadvantaged neighborhoods experience worse health, suggesting that exposure to crime could be a contributing factor. (Booth et al., 2014). The fact that income and crime density do not fully match shows that crime is shaped by a combination of social, economic, and environmental factors including land use, density, transit and commercial activity (Andresen & Linning, 2012).

Strengths and Limitations

This study had several strengths. It used a large, complete crime dataset that allowed for a detailed analysis. The GIS methods used were appropriate and helped make the mapping patterns easy to understand. Analyzing data at the neighborhood level was useful for practical planning purposes. Calculating crime by area made it easier to fairly compare neighborhoods. The maps were designed clearly and followed all required elements.

There were also limitations. The cross-sectional design means that the study cannot prove cause and effect (Wang & Cheng, 2020). Income data and crime data came from different years even though neighborhood income generally changes slowly. Seven neighborhoods had missing income data. Because the study looked at neighborhoods rather than individuals, ecological fallacy is a risk (Hsieh, 2016). Crime records include only reported crimes, and some communities report less than others (Skogan, 1977). The study also did not include other important variables such as land use, demographic details or transit access

Conclusions and Recommendations

This study shows large differences in crime exposure across Toronto neighborhoods, which has important public health implications. Crime density is highest in the downtown core and decreases as neighborhoods get farther from the city center. Understanding these patterns is important for city planning and decision-making. Public health services should consider crime exposure when choosing where to place services or programs. Neighborhoods with high crime density may need more mental health support, trauma care, or community programs focused on preventing violence. Crime density should also be included in health assessments and equity studies.

Future studies should include additional neighborhood-level variables such as demographic, land-use diversity, access to public transit and availability of social services. These variables may offer a more complete understanding of what causes crime patterns. Calculating crime per population (per capita) would provide a good perspective on crime exposure and individual risk. Finally, qualitative research that explores how residents perceive crime, experience victimization, and cope with neighborhood violence would improve our understanding and help guide interventions.

REFERENCES

Wang, X., & Cheng, Z. (2020). Cross-Sectional Studies: Strengths, Weaknesses, and Recommendations. *Chest*, *158*(1S), S65–S71. https://doi.org/10.1016/j.chest.2020.03.012

Hsieh, J. J. (2016, August 4). *Ecological fallacy | Epidemiology & Data Analysis*. Encyclopedia Britannica. https://www.britannica.com/science/ecological-fallacy

Andresen, M. A. (2006). Crime measures and the spatial analysis of criminal activity. *British Journal of Criminology, 46*(2), 258–285. https://doi.org/10.1093/bjc/azi054

Andresen, M. A., & Linning, S. J. (2012). The (in)appropriateness of aggregating across crime types. *Applied Geography, 35*(1–2), 275–282. https://doi.org/10.1016/j.apgeog.2012.07.007

Booth, G. L., Creatore, M. I., Gozdyra, P., Glazier, R. H., Tynan, A. M., & Booth, S. L. (2014). Neighbourhood walkability and the incidence of diabetes: An inverse probability of treatment weighting analysis. *Journal of Epidemiology and Community Health, 68*(11), 1109–1115. https://doi.org/10.1136/jech-2014-204399

Bursik, R. J., & Grasmick, H. G. (1993). *Neighborhoods and crime: The dimensions of effective community control*. Lexington Books.

Chainey, S., & Ratcliffe, J. (2005). *GIS and crime mapping*. John Wiley & Sons.

Cisler, J. M., Begle, A. M., Amstadter, A. B., Resnick, H. S., Danielson, C. K., Saunders, B. E., & Kilpatrick, D. G. (2012). Exposure to interpersonal violence and risk for PTSD, depression, delinquency, and binge drinking among adolescents: Data from the NSA-R. *Journal of Traumatic Stress, 25*(1), 33–40. https://doi.org/10.1002/jts.21672

City of Toronto. (2021). *Neighbourhoods* [Shapefile]. City of Toronto Open Data Portal. https://open.toronto.ca/dataset/neighbourhoods/

Curry, A., Latkin, C., & Davey-Rothwell, M. (2008). Pathways to depression: The impact of neighborhood violent crime on inner-city residents in Baltimore, Maryland, USA. *Social Science & Medicine, 67*(1), 23–30. https://doi.org/10.1016/j.socscimed.2008.03.007

Fitzgerald, R. T., Wisener, M., & Savoie, J. (2004). *Neighbourhood characteristics and the distribution of crime in Winnipeg* (Crime and Justice Research Paper Series). Statistics Canada.

Foster, S., & Giles-Corti, B. (2008). The built environment, neighborhood crime and constrained physical activity: An exploration of inconsistent findings. *Preventive Medicine, 47*(3), 241–251. https://doi.org/10.1016/j.ypmed.2008.03.017

Fowler, P. J., Tompsett, C. J., Braciszewski, J. M., Jacques-Tiura, A. J., & Baltes, B. B. (2009). Community violence: A meta-analysis on the effect of exposure and mental health outcomes of children and adolescents. *Development and Psychopathology, 21*(1), 227–259. https://doi.org/10.1017/S0954579409000145

Jenks, G. F., & Caspall, F. C. (1971). Error on choroplethic maps: Definition, measurement, reduction. *Annals of the Association of American Geographers, 61*(2), 217–244. https://doi.org/10.1111/j.1467-8306.1971.tb00779.x

Kawachi, I., Kennedy, B. P., & Wilkinson, R. G. (1999). Crime: Social disorganization and relative deprivation. *Social Science & Medicine, 48*(6), 719–731. https://doi.org/10.1016/S0277-9536(98)00400-6

Lorenc, T., Clayton, S., Neary, D., Whitehead, M., Petticrew, M., Thomson, H., Cummins, S., Sowden, A., & Renton, A. (2012). Crime, fear of crime, environment, and mental health and wellbeing: Mapping review of theories and causal pathways. *Health & Place, 18*(4), 757–765. https://doi.org/10.1016/j.healthplace.2012.04.001

Mair, C., Roux, A. V. D., & Galea, S. (2010). Are neighbourhood characteristics associated with depressive symptoms? A review of evidence. *Journal of Epidemiology and Community Health, 62*(11), 940–946. https://doi.org/10.1136/jech.2007.066605

Sampson, R. J., Raudenbush, S. W., & Earls, F. (1997). Neighborhoods and violent crime: A multilevel study of collective efficacy. *Science, 277*(5328), 918–924. https://doi.org/10.1126/science.277.5328.918

Sharkey, P. T., Tirado-Strayer, N., Papachristos, A. V., & Raver, C. C. (2012). The effect of local violence on children's attention and impulse control. *American Journal of Public Health, 102*(12), 2287–2293. https://doi.org/10.2105/AJPH.2012.300789

Silver, J. (2011). *Spatialized poverty in Winnipeg's inner city: The troubling case of the Lord Selkirk Park housing development*. Canadian Centre for Policy Alternatives Manitoba.

Skogan, W. G. (1977). Dimensions of the dark figure of unreported crime. *Crime & Delinquency, 23*(1), 41–50. https://doi.org/10.1177/001112877702300104

Statistics Canada. (2022). *Census Profile, 2021 Census of Population*. https://www12.statcan.gc.ca/census-recensement/2021/dp-pd/prof/index.cfm

Sundquist, K., Theobald, H., Yang, M., Li, X., Johansson, S. E., & Sundquist, J. (2006). Neighborhood violent crime and unemployment increase the risk of coronary heart disease: A multilevel study in an urban setting. *Social Science & Medicine, 62*(8), 2061–2071. https://doi.org/10.1016/j.socscimed.2005.08.051

Toronto Police Service. (2023). *Major Crime Indicators Open Data*. Toronto Police Service Public Safety Data Portal. https://data.torontopolice.on.ca/

Weisburd, D. (2015). The law of crime concentration and the criminology of place. *Criminology, 53*(2), 133–157. https://doi.org/10.1111/1745-9125.12070